

Differential Geometry and Lie Groups

Backpaper Examination

December 29, 2025

Instructions: All questions carry ten marks. All manifolds are assumed to be smooth.

1. State and prove the Mean Value Theorem for a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.
2. Let $S_0 = \{(x, y, z) \mid x^2 + y^2 = 1, |z| < 1\}$ denote the open cylinder in $\mathbb{R}^3$. Prove that it is a smooth oriented surface and find its tangent space and the normal direction at $p = (1, 0, 0)$.
3. Define the Gauss map and the curvature of an oriented surface $S \in \mathbb{R}^3$. Compute the Gauss map of the unit sphere in $\mathbb{R}^3$ and determine its curvature.
4. Let $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a smooth function and S denote the regular surface in $\mathbb{R}^3$ given by its graph. Prove that it is an orientable surface and show that the Gaussian curvature of S is given by the formula

$$K = \frac{h_{xx}h_{yy} - h_{xy}^2}{(1 + h_x^2 + h_y^2)^2}$$

where x, y are the coordinates of $\mathbb{R}^2$.

5. Prove that the set of invertible matrices of order n is a smooth manifold (under the induced topology from $\mathbb{R}^{n^2}$) and compute its tangent space at the identity matrix.
6. Let M be a manifold and X be a smooth vector field on M . Define integral curve for X . Prove that the domain of the integral curve of a left-invariant vector field X on a Lie Group G is the whole of the real line (i.e. any such X is a *complete* vector field).

7. Define a C^∞ -connection ∇ on a manifold M . If X, Y are smooth vector fields on M then for any $p \in M$, prove that $\nabla_X Y|_p$ depends on Y near p and X at p .
8. Define the curvature tensor on M . If S is a surface in $\mathbb{R}^3$, then express the Gaussian curvature of S at a point p in terms of the curvature tensor of the Riemannian connection associated with its First Fundamental form.
9. Define sectional curvature of a manifold M at a point p . Prove that the sectional curvature is well defined.
10. Let M be a manifold and $p \in M$ be a point. Define the Ricci curvature of M at p . Prove that it is well defined.